

James Notley

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Skills

Fabrication: UHV systems, CNC machining, CFRP composites, FDM 3D printing, sheet metal

Characterization: universal testing machine, metallography, SEM, XRD

CAD: SolidWorks, OnShape, Fusion360+simulation, Creo

Code: Python, bash, VBA, SQL

Experience

Researcher (MEMENCYS Fellowship), Bartels Lab — Riverside, CA Jan 2026 – Present

- Designed and built a manipulator-compatible heated stage that allows silicon substrates annealed at 1000°C as they move through different stages of a semiconductor material growth and metallization process under ultra high vacuum
- Designed and built a vacuum chamber for deposition of up to three separate metals in one process step
- Implemented and qualified a thermal regulation system for the deposition rate measurement device in the metal deposition chamber to reduce thermal drift such that layer thickness error is <5nm

Airframe Lead, Highlander Space Program – Riverside, CA Oct 2023 – June 2026

- Led a team of 7 students in the design and fabrication of two launch vehicle airframes, one first place in competition
- Simulated flight based on engine performance using OpenRocket and optimized to hit a contract altitude $\pm 10\%$
- Verified structural performance through simulations and hand calcs, and optimized for weight still maintaining FOS>2
- Fabricated airframes from fiberglass and sheet metal entirely in-house

Manufacturing Engineering Intern, Woodward HRT – Santa Clarita, CA June 2025 – Sept 2025

- Implemented data exchange between tool measurement and CNC machining centers, eliminating manual measurement and input error and saving up to \$20k/y per machine in scrap reduction
- Built a VBA tool to detect and report discrepancies between Sharepoint and SAP ERPs with one click

Machinist Intern, S and S Machine – Roseville, CA June 2024 – Sept 2024

- Identified and solved issues with CAM-generated toolpaths not caught by software or other machinists, such as strange lifting during a finishing pass caused by excessive tool engagement on pocket corners and fixed by pre-drilling
- Inspected first articles to GD&T standards using both CMM and traditional inspection methods

Projects

Metal evaporation chamber (Senior design poster) raptor8134.github.io/pdf/evap_chamber.pdf

Composite filament winder raptor8134.github.io/projects/filament-winder.html

- Built a filament winding machine to fabricate fiberglass tubes and nosecones for rocketry at costs up to 90% under market price for under \$500 with mostly 3D printed parts and no custom electronics
- Wrote toolpath software with capabilities for complex mandrel geometries (nosecones) and an intuitive interface
- Prototyped a resin applicator with zero excess resin drip, removing the need for expensive prepreg and reducing mess

Education

UC Riverside – BS in Materials Science and Engineering

Dec 2026

3.71 GPA, Tau Beta Pi